

Mathematical Explanation of Data Re-Uploading in Variational Quantum Circuits

1 Problem Setting

In the `sin_encoded` experiment, the target function is

$$y = \sin(2\pi x) + \varepsilon, \quad x \in [0, 1], \quad \varepsilon \sim \mathcal{N}(0, \sigma^2). \quad (1)$$

The model is a variational quantum circuit (VQC) whose prediction is an expectation value of a measured observable,

$$f_{\boldsymbol{\theta}}(x) = \langle 0|U^\dagger(x, \boldsymbol{\theta})OU(x, \boldsymbol{\theta})|0\rangle, \quad (2)$$

where O is a Pauli observable and $\boldsymbol{\theta}$ are trainable parameters. Since Pauli expectation values lie in $[-1, 1]$, this output range is well matched to a sine-regression target.

2 Angle Encoding

A common way to load a classical scalar x into a quantum circuit is angle encoding. For one qubit, the feature map can be written as

$$S(x) = R_Y(x), \quad R_Y(x) = \exp\left(-i\frac{x}{2}Y\right). \quad (3)$$

After this rotation, the quantum state contains trigonometric dependence on the input:

$$R_Y(x)|0\rangle = \cos\left(\frac{x}{2}\right)|0\rangle + \sin\left(\frac{x}{2}\right)|1\rangle. \quad (4)$$

Therefore, any observable expectation built from this encoded state is a trigonometric function of x . However, if the input is inserted only once and at one frequency scale, the accessible Fourier spectrum is limited [2].

3 Proof of Spectral Limitation with Single Encoding

To understand why a single encoding is insufficient for complex regression tasks, we prove that an architecture with only one data-loading operation is strictly bounded to the fundamental frequency of the input.

Goal: Prove that a VQC utilizing a single angle encoding operation generates an output strictly constrained to the base frequency, regardless of the subsequent trainable operations.

Assumptions & Prerequisites:

- A single-qubit system initialized in the state $|0\rangle$.
- An encoding feature map $S(x) = R_Y(x)$.
- Measurement via the Pauli-Z observable $O = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

- For simplicity, we isolate the encoding by measuring immediately, abstracting away trainable parameters which can only scale or shift amplitudes but cannot fundamentally alter the trigonometric arguments.

Derivation:

The state vector (ket) after the encoding gate is applied is:

$$|\psi\rangle = \begin{bmatrix} \cos\left(\frac{x}{2}\right) \\ \sin\left(\frac{x}{2}\right) \end{bmatrix}$$

The corresponding dual vector (bra) is:

$$\langle\psi| = \begin{bmatrix} \cos\left(\frac{x}{2}\right) & \sin\left(\frac{x}{2}\right) \end{bmatrix}$$

The expectation value is the scalar result of the inner product $\langle\psi|Z|\psi\rangle$:

$$f(x) = \begin{bmatrix} \cos\left(\frac{x}{2}\right) & \sin\left(\frac{x}{2}\right) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos\left(\frac{x}{2}\right) \\ \sin\left(\frac{x}{2}\right) \end{bmatrix}$$

First, applying the observable Z to the ket:

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos\left(\frac{x}{2}\right) \\ \sin\left(\frac{x}{2}\right) \end{bmatrix} = \begin{bmatrix} \cos\left(\frac{x}{2}\right) \\ -\sin\left(\frac{x}{2}\right) \end{bmatrix}$$

Next, multiplying the bra by the resulting vector:

$$\begin{aligned} f(x) &= \cos\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right) \sin\left(\frac{x}{2}\right) \\ f(x) &= \cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right) \end{aligned}$$

Applying the double-angle trigonometric identity $\cos(2\alpha) = \cos^2(\alpha) - \sin^2(\alpha)$, we obtain the final prediction:

$$f(x) = \cos(x)$$

Conclusion:

The algebraic projection during measurement strictly halves the geometry of the Bloch sphere, converting the $x/2$ angles inside the state vector into an observable output of $\cos(x)$. Because linear quantum operations cannot raise the power of these trigonometric terms prior to measurement, the model is trapped at this fundamental frequency.

4 Data Re-Uploading

Data re-uploading increases expressivity by inserting the same classical input multiple times into the circuit, usually interleaved with trainable blocks [1]. A generic data re-uploading circuit has the form

$$U(x, \boldsymbol{\theta}) = W_L(\boldsymbol{\theta}_L) S_L(x) \cdots W_2(\boldsymbol{\theta}_2) S_2(x) W_1(\boldsymbol{\theta}_1) S_1(x), \quad (5)$$

where each $S_\ell(x)$ is a data-dependent encoding block and each $W_\ell(\boldsymbol{\theta}_\ell)$ is a trainable unitary. Repeatedly encoding the same input lets the trainable circuit combine several nonlinear trigonometric components.

In the `sin_encoded` notebook, the re-uploading is implemented across four qubits by first scaling the scalar input and then assigning one scaled copy to each qubit:

$$\tilde{x} = \text{MinMaxScale}(x) \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right], \quad (6)$$

$$\mathbf{z}(x) = (\tilde{x}, 2\tilde{x}, 3\tilde{x}, 4\tilde{x}). \quad (7)$$

The feature map is then

$$S(x) = \bigotimes_{k=1}^4 R_Y(k\tilde{x}). \quad (8)$$

Equivalently, qubit k receives the rotation angle $k\tilde{x}$. This creates several Fourier components of the same original scalar input.

5 Proof of Enhanced Expressivity via Parallel Re-Uploading and Entanglement

We now prove that parallel data re-uploading across multiple qubits, when subjected to quantum entanglement, mathematically generates novel frequency harmonics that did not exist in the initial feature map.

Goal: Demonstrate that combining different input frequencies via a trainable ansatz and subsequent measurement forces the emergence of higher-order trigonometric mixed terms, increasing the model's capacity.

Assumptions & Prerequisites:

- A 2-qubit system initialized in $|00\rangle$.
- Parallel angle encoding where Qubit 1 receives x and Qubit 2 receives a scaled input ax (where $a \in \mathbb{R}$).
- A trainable ansatz layer capable of multi-qubit entanglement.
- Measurement via the composite observable $O = Z \otimes Z$.

Derivation:

We begin by encoding the data independently onto the two qubits. The initial state is the tensor product of the two individual states:

$$|\psi_{\text{enc}}\rangle = \left(\cos\left(\frac{x}{2}\right)|0\rangle + \sin\left(\frac{x}{2}\right)|1\rangle \right) \otimes \left(\cos\left(\frac{ax}{2}\right)|0\rangle + \sin\left(\frac{ax}{2}\right)|1\rangle \right)$$

Expanding this tensor product reveals the initial mixed amplitudes. Let us isolate the amplitude corresponding to the $|00\rangle$ basis state:

$$\alpha_{00}^{\text{init}} = \cos\left(\frac{x}{2}\right) \cos\left(\frac{ax}{2}\right)$$

Using the product-to-sum trigonometric identity $\cos(\alpha)\cos(\beta) = \frac{1}{2}[\cos(\alpha - \beta) + \cos(\alpha + \beta)]$, we rewrite this amplitude:

$$\alpha_{00}^{\text{init}} = \frac{1}{2} \left[\cos\left(\frac{(1-a)x}{2}\right) + \cos\left(\frac{(1+a)x}{2}\right) \right]$$

To simplify the subsequent algebra, let $A = \cos\left(\frac{(1-a)x}{2}\right)$ and $B = \cos\left(\frac{(1+a)x}{2}\right)$. The entangling ansatz now performs linear combinations of the system's amplitudes. The new amplitude for the $|00\rangle$ state becomes a weighted sum influenced by the trainable parameters θ :

$$\alpha_{00} = w_1 A + w_2 B$$

To calculate the expected value during measurement, we must compute the squared modulus of this amplitude, $|\alpha_{00}|^2$. It is in the squaring of this entire sum that quantum interference mathematically manifests:

$$|\alpha_{00}|^2 = (w_1 A + w_2 B)^2 = w_1^2 A^2 + w_2^2 B^2 + 2w_1 w_2 AB$$

Let us evaluate the crucial cross-term, $2AB$, which represents the interference between the states:

$$2AB = 2 \cos\left(\frac{(1-a)x}{2}\right) \cos\left(\frac{(1+a)x}{2}\right)$$

Applying the product-to-sum identity once more to the $2AB$ term:

$$2AB = \cos(ax) + \cos(x)$$

Simultaneously, the squared terms A^2 and B^2 evaluate to base harmonic components via half-angle identities (e.g., $A^2 = \frac{1+\cos((1-a)x)}{2}$).

Conclusion:

The resulting expectation value aggregates these components into a finite Fourier series containing frequencies $\{1, a, 1-a, 1+a\}$. The $2AB$ cross-term proves that as long as the optimizer maintains non-zero weights in the ansatz ($w_1 w_2 \neq 0$), the original frequencies x and ax survive and combine with the newly generated sums and differences. Thus, parallel data re-uploading strictly increases the spectral expressivity of the model.

6 Fourier View

The reason data re-uploading helps is that expectation values of parametrized quantum circuits with angle encodings can be represented as finite Fourier series in the input [2]. Abstractly,

$$f_{\boldsymbol{\theta}}(x) = \sum_{\omega \in \Omega} c_{\omega}(\boldsymbol{\theta}) e^{i\omega x}, \quad (9)$$

or, for real-valued outputs,

$$f_{\boldsymbol{\theta}}(x) = a_0(\boldsymbol{\theta}) + \sum_{\omega \in \Omega_+} [a_{\omega}(\boldsymbol{\theta}) \cos(\omega x) + b_{\omega}(\boldsymbol{\theta}) \sin(\omega x)]. \quad (10)$$

The set Ω of available frequencies is determined by how the input enters the circuit. With only a single encoding $R_Y(\tilde{x})$, the model can express only a small Fourier family. With the four re-uploaded angles $\tilde{x}, 2\tilde{x}, 3\tilde{x}, 4\tilde{x}$, the model has access to a richer frequency set. The trainable ansatz and entangling gates then learn the coefficients $a_{\omega}(\boldsymbol{\theta})$ and $b_{\omega}(\boldsymbol{\theta})$.

This is the quantum analogue of replacing a scalar feature x with classical Fourier features such as

$$\phi(x) = (\sin x, \cos x, \sin 2x, \cos 2x, \sin 3x, \cos 3x, \sin 4x, \cos 4x). \quad (11)$$

The circuit does not explicitly construct this classical vector. Instead, the feature map and subsequent quantum operations implicitly generate and combine these trigonometric components.

7 Why It Helps for Sine Regression

The target function $\sin(2\pi x)$ is itself a Fourier mode. A circuit that sees the input only through one weak angle encoding may not have enough spectral bandwidth to approximate the target accurately after scaling and noise are included. Data re-uploading expands the available basis functions, so the model can fit the dominant sine mode and use additional components to absorb finite-sample and noisy-data effects.

For the four-qubit encoding used in `sin_encoded`, the model input is not four independent features. It is one scalar x represented at four different frequency scales. The improvement therefore comes from a richer encoding of the same information, not from adding extra data.

8 Conclusion

Data re-uploading makes the VQC more expressive by allowing the measured expectation value to contain a richer Fourier expansion of the input. In the `sin_encoded` experiment, the transformation

$$x \mapsto (\tilde{x}, 2\tilde{x}, 3\tilde{x}, 4\tilde{x})$$

is mathematically appropriate for a sine-regression task because it gives the quantum model access to multiple trigonometric frequencies of the same scalar input.

References

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